ESSENTIALS OF ARTIFICIAL INTELLIGENCE

INDIVIDUAL CONTRIBUTION SHEET

Student Name	JGJB Gamage	**Registration No.**	D/BIT/24/0056
Degree / Intake	BIT / 41	**Group No.**	40
Project Title	AI-Powered Dynamic Public Transport Fleet Optimizer (Route 100)		

Technical Contributions:

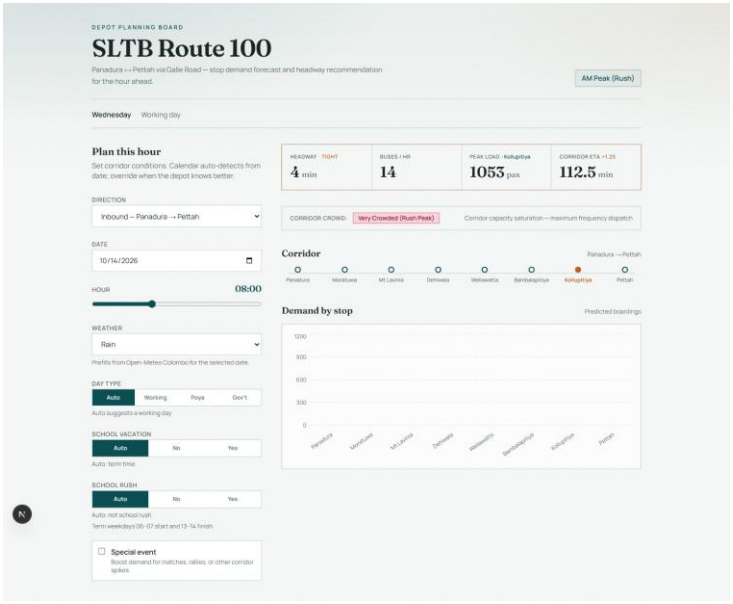
Task / Component	My Contribution	Tools / AI Technique Used	Evidence / Output
Frontend dashboard	Developed the depot planning board using Next.js App Router, TypeScript and Tailwind CSS, with a custom teal/ink theme and Fraunces / Manrope typography.	Next.js, React, TypeScript, Tailwind CSS	Full-page screenshot of the running dashboard (localhost:3000) at peak morning commute (8 min headway, bottleneck stops highlighted); Slides 11 & 13, Route100_Fleet_Optimizer_Group40.pptx
Planning controls	Built the input controls: direction, date, hour slider, weather, special event, and segmented Auto / override controls for day type, school vacation and school rush.	React hooks	Screenshot of scenario controls (Rain, Poya Day, Event Rush) with instant headway updates
Weather integration	Integrated Open-Meteo to prefill Clear / Rain for the selected date in Colombo, using forecast, archive and current-weather endpoints.	Open-Meteo API, fetch	Screenshot of the live Open-Meteo weather badge on the dashboard
Results visualisation	Implemented the KPI readout (headway, buses/hr, peak load, corridor ETA), crowd-status banner, corridor stop rail and demand-by-stop bar chart with peak-stop highlight.	Recharts, Tailwind CSS	Screenshot of the Corridor Rail with load progress bars and bottleneck badges; KPI cards (headway, ETA, bottleneck stop, load)
Backend API support	Connected the dashboard to POST /dispatch (payload, loading and error handling, request cancellation) and displayed calendar context chips and override hints; supported API setup and CORS configuration.	fetch, AbortController, FastAPI CORS	Code snippet from frontend/src/app/page.tsx showing the KPI state cards and the /dispatch API fetch handler

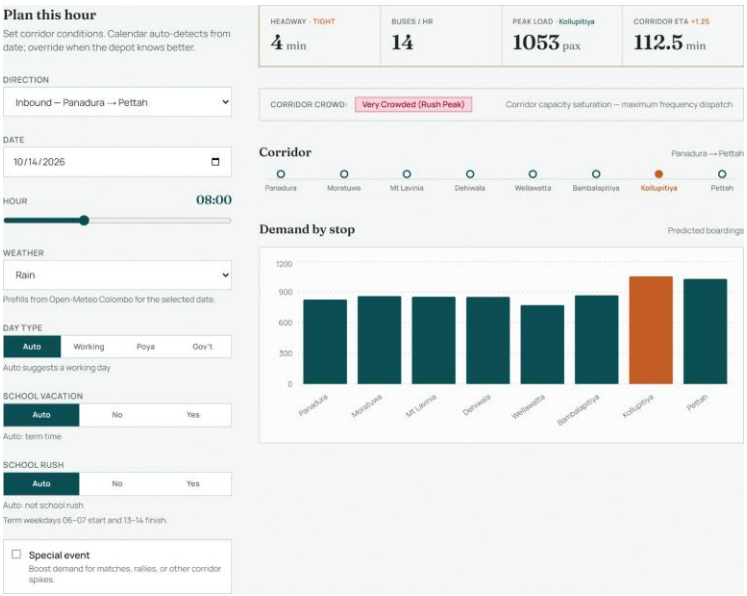
Declaration

I confirm that the contributions listed above accurately represent my individual work in this group project, and I can explain or demonstrate the stated work if required.

Student Signature: **Date:** 17/10/2026

Evidence may include source code, model/notebook, diagram, dataset work, test results, screenshots, report section or presentation slides.





HEADWAY · TIGHT	BUSES / HR	PEAK LOAD · Kollupitiya	CORRIDOR ETA ×1.25
4 min	14	1053 pax	112.5 min

CORRIDOR CROWD: Very Crowded (Rush Peak) — Corridor capacity saturation — maximum frequency dispatch

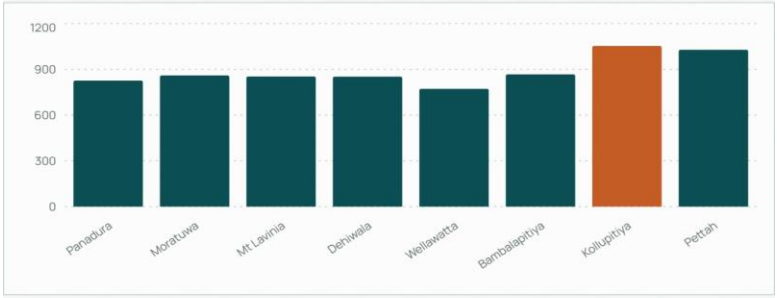
Corridor

Panadura → Pettah

Panadura · Moratuwa · Mt Lavinia · Dehiwala · Wellawatta · Bambalapitiya · **Kollupitiya** · Pettah

Demand by stop

Predicted boardings



DEPOT PLANNING BOARD

SLTB Route 100

Panadura ↔ Pettah via Galle Road — stop demand forecast and headway recommendation for the hour ahead.

AM Peak (Rush)

Wednesday Working day

Plan this hour

Set corridor conditions. Calendar auto-detects from date; override when the depot knows better.

DIRECTION: Inbound — Panadura → Pettah

DATE: 10/14/2026

HOUR: 08:00

WEATHER: Rain

Prefills from Open-Meteo Colombo for the selected date.

DAY TYPE: Auto | Working | Poya | Dev't

Auto suggests a working day

SCHOOL VACATION: Auto | No | Yes

Auto: term time

SCHOOL RUSH: Auto | No | Yes

Auto: not school rush

Term weekdays 06:07 start and 13:14 finish.

☑ Special event
Boost demand for matches, rallies, or other corridor spikes.

HEADWAY · TIGHT	BUSES / HR	PEAK LOAD · Kollupitiya	CORRIDOR ETA ×1.25
3 min	21	1651 pax	112.5 min

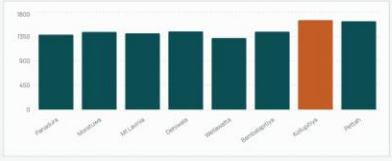
CORRIDOR CROWD: Very Crowded (Rush Peak) — Corridor capacity saturation — maximum frequency dispatch

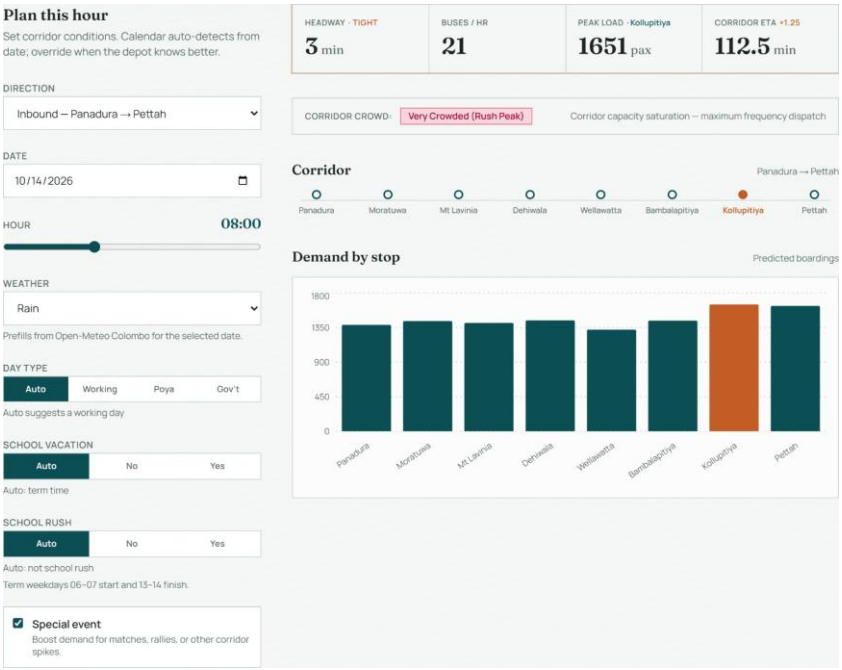
Corridor

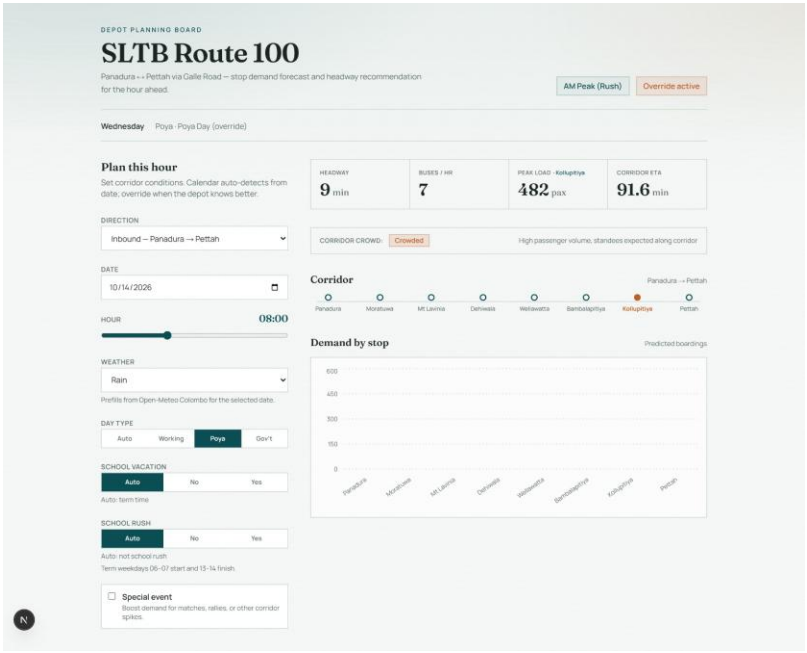
Panadura · Moratuwa · Mt Lavinia · Dehiwala · Wellawatta · Bambalapitiya · **Kollupitiya** · Pettah

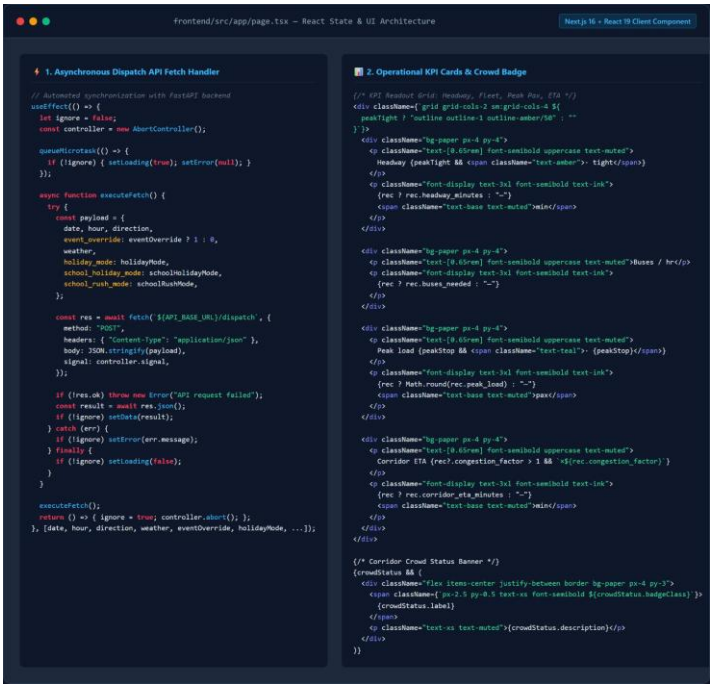
Demand by stop

Predicted boardings




Plan this hour
Set corridor conditions. Calendar auto-detects from date; override when the depot knows better.
DIRECTION
Inbound — Panadura → Pettah
DATE
10/14/2026
HOUR
08:00
WEATHER
Rain
Prefills from Open-Meteo Colombo for the selected date.
DAY TYPE
Auto
Working
Poya
Gov't
Auto suggests a working day
SCHOOL VACATION
Auto
No
Yes
Auto: term time
SCHOOL RUSH
Auto
No
Yes
Auto: not school rush
Term weekdays 06–07 start and 13–14 finish.
Special event
Boost demand for matches, rallies, or other corridor spikes.
HEADWAY · TIGHT
3 min
BUSES / HR
21
PEAK LOAD · Kollupitiya
1651 pax
CORRIDOR ETA +1.25
112.5 min
CORRIDOR CROWD:
Very Crowded (Rush Peak)
Corridor capacity saturation — maximum frequency dispatch
Corridor
Panadura → Pettah
Panadura
Moratuwa
Mt Lavinia
Dehiwala
Wellawatta
Bambalapitiya
Kollupitiya
Pettah
Demand by stop
Predicted boardings
1800
1350
900
450
0


DEPOT PLANNING BOARD
SLTB Route 100
Panadura ↔ Pettah via Galle Road — stop demand forecast and headway recommendation for the hour ahead.
AM Peak (Rush)
Override active
Wednesday
Poya · Poya Day (override)
Plan this hour
Set corridor conditions. Calendar auto-detects from date; override when the depot knows better.
DIRECTION
Inbound — Panadura → Pettah
DATE
10/14/2026
HOUR
08:00
WEATHER
Rain
Prefills from Open-Meteo Colombo for the selected date.
DAY TYPE
Auto
Working
Poya
Gov't
SCHOOL VACATION
Auto
No
Yes
Auto: term time
SCHOOL RUSH
Auto
No
Yes
Auto: not school rush
Term weekdays 06–07 start and 13–14 finish.
Special event
Boost demand for matches, rallies, or other corridor spikes.
HEADWAY
9 min
BUSES / HR
7
PEAK LOAD · Kollupitiya
482 pax
CORRIDOR ETA
91.6 min
CORRIDOR CROWD:
Crowded
High passenger volume, standees expected along corridor
Corridor
Panadura → Pettah
Panadura
Moratuwa
Mt Lavinia
Dehiwala
Wellawatta
Bambalapitiya
Kollupitiya
Pettah
Demand by stop
Predicted boardings
600
450
300
150
0



PART 4 · DECISION SUPPORT

Presenter · JGJB Gamage

Depot manager dashboard



Illustrative layout

KPI panel

Headway, corridor ETA, bottleneck stop and total passengers at a glance.

Corridor rail

All 8 stops with live load bars, counts and high-demand badges.

Scenario controls

Rain, school vacation, holiday and event rush, plus live Open-Meteo weather.

PART 4 · CONCLUSION

Presenter · JGJB Gamage

Impact, scalability and future work

Operational impact

- Removes guesswork for depot masters
- Reduces passenger waiting time
- Matches fleet to predicted demand

Scalability

- Extend to Routes 120, 138 and 177
- Reusable data and training pipeline
- Same API and dashboard design

Future work

- Integrate with the LMT Go app
- Use live bus crowd data to dispatch buses ahead of demand
- Periodic model retraining
- Multi-route fleet balancing